

COOLING SYSTEMS AND POWER MANAGEMENT

1. Measure your CPU temperature at idle and under load using a tool like HWMonitor or Core Temp, then record the readings before and after applying new thermal paste and reinstalling your CPU cooler.

-> CPU Temperature Before and After Reapplying Thermal Paste

Use **HWMonitor** or **Core Temp** to record the CPU temperature once at idle and again during a CPU load test. Then power off the system, reapply thermal paste, reinstall the cooler, and repeat the same test under similar room and workload conditions.

Test condition	Before new thermal paste	After new thermal paste
Idle temperature	47°C	39°C
Load temperature	82°C	71°C
Temperature improvement	—	11°C lower under load

Example observation

Before replacing the thermal paste, the CPU temperature was approximately **47°C at idle** and **82°C under load**. After cleaning the old paste, applying a pea-sized amount of new thermal paste, and reinstalling the cooler evenly, the temperature decreased to approximately **39°C at idle** and **71°C under load**.

Conclusion

The CPU temperatures were lower after applying fresh thermal paste and reseating the cooler. The biggest improvement was observed under load, where the example temperature decreased by **11°C**, indicating better heat transfer from the CPU to the cooler.

Important: These are **example readings for your lab record**, not measurements from your actual PC. Replace them with the values shown by HWMonitor/Core Temp on your system.

2. Write a step-by-step guide (with photos or diagrams) showing how to correctly apply thermal paste and mount a CPU cooler for optimal heat transfer.

Hint: Include tips on how much paste to use, spreading methods, and common mistakes to avoid.

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How to Apply Thermal Paste and Mount a CPU Cooler

Better cooling = lower temperatures = better performance = longer CPU life

1 Gather your materials

- CPU cooler (with mounting kit)
- Thermal paste
- Screwdriver
- Isopropyl alcohol + lint-free cloth
- (Optional) Gloves

2 Remove old paste (if any)

- Gently clean the CPU heat spreader and cooler base with isopropyl alcohol and a lint-free cloth.
- Make sure there is no old paste or dust.

3 Apply thermal paste

- Use a pea-sized dot (about 0.1–0.2 ml) for most CPUs.
- Do not overuse – too much paste can reduce cooling performance.

4 Spread the paste (optional)

- You can either spread it with a clean card/plastic tool or let the cooler spread it when you mount it.
- The goal is a thin, even layer (not too thick).

5 Mount the cooler

- Align the cooler with the mounting holes and place it gently on the CPU.
- Make sure it sits flat and centered.

6 Secure the cooler

- Use the correct mounting hardware for your cooler (screws, brackets, or push pins).
- Tighten screws gradually in a cross pattern (e.g., top-left → bottom-right → top-right → bottom-left) to ensure even pressure.

7 Connect the fan

- Plug the cooler fan into the CPU_FAN header on the motherboard.
- Make sure the cable is fully seated.

8 Final check

- Check that the cooler is firmly mounted, the fan is connected, and there are no loose cables.
- Reconnect power and boot the system.

9 Test temperatures

CPU	
Package	45 °C (113 °F)
Core #0	43 °C (109 °F)
Core #1	44 °C (111 °F)

- Use HWMonitor, Core Temp or BIOS to check temperatures at idle and under load.
- Lower temps = better thermal contact.

How much thermal paste to use?

Too little

Just right

Too much

Spreading methods (choose one)

1. Pea-sized dot (recommended)
2. Thin layer (spread with card/tool)
3. X pattern (alternative)

All three methods work. The most important thing is a thin, even layer.

Common mistakes to avoid

- ✗ Using too much or too little paste.
- ✗ Spreading paste with your finger (can introduce oils and dirt).
- ✗ Not cleaning old paste properly.
- ✗ Mounting the cooler unevenly (causes poor contact).
- ✗ Forgetting to connect the fan cable.
- ✗ Using the wrong mounting hardware (check your cooler's manual).

Pro Tips

- Always use high-quality thermal paste (e.g., Arctic MX-4, Noctua NT-H1, etc.).
- Keep your workspace clean and static-free.
- Check your cooler's manual for specific mounting instructions.
- Reapply thermal paste every 1–2 years or if you remove the cooler.

Good thermal contact = cooler CPU = better performance!

How to Apply Thermal Paste and Mount a CPU Cooler

The image above provides the requested **photo/diagram documentation**. Use this step-by-step procedure alongside it.

1. Prepare the components

Shut down and unplug the PC. Gather the CPU cooler, thermal paste, screwdriver, and a lint-free cloth with isopropyl alcohol.

2. Clean the CPU

Remove any old thermal paste from the CPU heat spreader and cooler contact surface. Allow the surfaces to dry completely.

3. Apply thermal paste

Place **one small pea-sized dot** of thermal paste in the center of the CPU. For most desktop CPUs, approximately **0.1–0.2 mL** is sufficient.

4. Choose a spreading method

The preferred simple method is the **pea-sized dot** and allowing the cooler's mounting pressure to spread it. A thin, even layer using a clean plastic applicator can also work when recommended by the cooler/paste manufacturer.

5. Position the cooler

Align the cooler with the mounting holes or retention brackets and lower it **straight down** onto the CPU. Avoid sliding it around unnecessarily.

6. Secure the cooler evenly

Tighten the screws gradually in a **cross/diagonal pattern** so the mounting pressure is distributed evenly. Follow the cooler manufacturer's mounting instructions.

7. Connect the CPU fan

Plug the cooler's fan connector firmly into the motherboard's **CPU_FAN** header.

8. Perform a final check

Make sure the cooler is firmly mounted, the fan cable is connected, and no cables are touching the fan blades.

9. Check temperatures

Boot the computer and use BIOS, **HWiNFO**, **HWMonitor**, or **Core Temp** to check idle and load CPU temperatures.

Common mistakes to avoid

Mistake	Why it is a problem
Too much thermal paste	Can make a mess and may reduce heat-transfer effectiveness
Too little paste	May not adequately cover the CPU surface
Touching paste with your finger	Can introduce oils and contaminants
Leaving old paste on the CPU	Prevents good contact between surfaces
Mounting the cooler unevenly	Creates uneven pressure and poor thermal contact
Forgetting the CPU_FAN connection	Can cause fan/CPU warnings or overheating
Using incorrect mounting hardware	May prevent proper cooler contact

3. Calculate the recommended PSU wattage for three different PC builds: a basic office PC, a gaming PC, and a content creator PC (with GPU and multiple drives). List all major components and show your calculations for each build.

-> Recommended PSU Wattage for Three PC Builds

A simple way to estimate PSU capacity is:

Recommended PSU wattage = Estimated component power × 1.3

The extra ~30% provides headroom for power spikes, future upgrades, and PSU efficiency.

1. Basic Office PC

Component	Estimated power
CPU – Intel Core i3 / Ryzen 3	65 W
Motherboard	50 W
RAM – 16 GB	10 W
500 GB SSD	5 W
Integrated graphics	0 W additional
2 case fans	6 W
USB/peripherals	15 W
Estimated total	151 W

Calculation:

151 W × 1.30 = **196.3 W**

Recommended PSU: 300 W

A 300 W quality PSU provides comfortable headroom for an office system and minor future upgrades.

2. Gaming PC

Component	Estimated power
CPU – Ryzen 5 / Core i5	90 W
GPU – mid-range gaming GPU	220 W
Motherboard	60 W

Component	Estimated power
RAM – 16 GB	10 W
1 TB NVMe SSD	7 W
2–3 case fans	10 W
CPU cooler	5 W
USB/peripherals	15 W
Estimated total	417 W

Calculation:

$$417 \text{ W} \times 1.30 = \mathbf{542.1 \text{ W}}$$

Recommended PSU: 650 W

A 650 W PSU gives additional capacity for GPU power spikes and future component upgrades.

3. Content Creator PC

This example includes a more powerful CPU, dedicated GPU, and multiple storage drives.

Component	Estimated power
CPU – high-performance Ryzen 9 / Core i9 class	170 W
GPU – high-end content-creation GPU	350 W
Motherboard	70 W
RAM – 32/64 GB	25 W
NVMe SSD #1	8 W
NVMe SSD #2	8 W
2 × HDD	16 W
4 case fans	12 W
CPU liquid/air cooler	10 W

Component	Estimated power
USB/peripherals	20 W
Estimated total	689 W

Calculation:

$$689 \text{ W} \times 1.30 = 895.7 \text{ W}$$

Recommended PSU: 1000 W

A 1000 W PSU provides useful headroom for a high-power GPU, multiple drives, sustained rendering workloads, and future upgrades.

Final Comparison

Build	Estimated load	30% headroom	Recommended PSU
Basic Office PC	151 W	196 W	300 W
Gaming PC	417 W	542 W	650 W
Content Creator PC	689 W	896 W	1000 W

4. Compare the total 5-year electricity cost for two PSUs: one rated 80 Plus Bronze (85% efficiency) and one rated 80 Plus Gold (90% efficiency), both running a 400W average load for 8 hours daily at ₹8 per kWh. Show your math and final cost difference.
Hint: Remember to account for efficiency when calculating actual power drawn from the wall.

-> 5-Year Electricity Cost: Bronze vs Gold PSU

The PC needs to receive **400 W** continuously while running. Because the PSU is not 100% efficient, it must draw more than 400 W from the wall.

1. 80 Plus Bronze — 85% efficiency

Wall power drawn:

$$400 \div 0.85 = 470.59W$$

Daily energy:

$$470.59 \times 8 = 3764.72Wh = 3.765kWh$$

5-year energy:

$$3.765 \times 365 \times 5 = 6,871.8kWh$$

5-year electricity cost:

$$6,871.8 \times ₹8 = ₹54,974.40$$

2. 80 Plus Gold — 90% efficiency

Wall power drawn:

$$400 \div 0.90 = 444.44W$$

Daily energy:

$$444.44 \times 8 = 3555.52Wh = 3.556kWh$$

5-year energy:

$$3.556 \times 365 \times 5 = 6,488.9kWh$$

5-year electricity cost:

$$6,488.9 \times ₹8 = ₹51,911.11$$

Final Comparison

PSU	Efficiency	Wall power	5-year electricity cost
80 Plus Bronze	85%	470.59 W	₹54,974.40
80 Plus Gold	90%	444.44 W	₹51,911.11
Difference	—	26.15 W	₹3,063.29

5. Create a table comparing air cooling vs liquid cooling for desktop PCs, listing at least 3 pros and 3 cons for each based on performance, noise, maintenance, and cost.

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Category	Air Cooling	Liquid Cooling (AIO)
Performance – Pros	• Good cooling for most CPUs • Large tower coolers can handle high loads • No pump to fail	• Can offer strong cooling, especially with high-end CPUs • Large radiator provides substantial heat-dissipation area • Often useful for sustained heavy workloads

Category	Air Cooling	Liquid Cooling (AIO)
Performance – Cons	• Very high-end CPUs may benefit from stronger cooling • Large heatsinks can interfere with RAM or case space • Cooling depends heavily on case airflow	• Performance depends on radiator size, airflow, and pump quality • Small radiators may perform similarly to good air coolers • Pump failure can eventually stop cooling
Noise – Pros	• No pump noise • Simple fan-based design • Can be very quiet with a large, slow fan	• Radiators can use multiple low-speed fans • Heat can be distributed efficiently • Some AIOs remain quiet at moderate loads
Noise – Cons	• Fans may become loud under heavy CPU loads • High-RPM fans can produce noticeable noise • Large heatsinks may require stronger airflow	• Includes both fan and pump noise • Pump can produce a constant hum or vibration • More fans can mean more potential noise
Maintenance – Pros	• Simple design • Usually only requires periodic dust cleaning • No liquid or pump to monitor	• Sealed AIOs generally require little routine maintenance • No need to refill most sealed units • Radiator can be cleaned like a case fan
Maintenance – Cons	• Dust can accumulate on the heatsink and fans • Thermal paste eventually needs replacement • Removing a large cooler can be inconvenient	• More components can potentially fail • Pump or tubing failure requires replacement • Radiator and fans can collect dust
Cost – Pros	• Generally cheaper • Excellent value for mainstream CPUs • Low-cost replacement parts	• Available across a wide range of prices • Can provide premium cooling for high-end builds • Often chosen for aesthetics as well as cooling
Cost – Cons	• Premium air coolers can become expensive • Very large coolers may require a larger case	• Usually more expensive than a comparable air cooler • Premium AIOs can be costly • Replacement cost after pump failure can be higher

